

REVIEW

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Examining methodological rigor, ethical governance, and scientific claims: a critical review of the DeepSeek-R1 study

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Abstract

The publication of DeepSeek-R1 in *Nature* marks a significant moment as the first widely recognized large language model to undergo formal peer review. While acknowledging its technical ambitions and contributions, this commentary provides a critical analysis focused on three core dimensions where the research falls short of the rigorous standards expected for high-impact scientific reporting. First, we examine methodological and transparency deficits, particularly concerning opaque training data disclosure and challenges in reproducibility. Second, we assess overlooked ethical and safety risks, including the implications of training data contamination, limited safety evaluations, and unresolved questions of accountability for AI-generated content. Third, we critically reappraise its scientific contributions, scrutinizing claims of innovation, cost-efficiency, and the peer review process itself. These interconnected issues, if unaddressed, risk undermining the credibility and cumulative progress of AI research. We conclude by proposing targeted, actionable recommendations—such as mandatory structured data documentation, interdisciplinary ethics review, and independent auditing—aimed at relevant stakeholders to foster a more robust, transparent, and responsible research ecosystem.

Keywords LLMs, DeepSeek-R1, Artificial intelligence, AI ethics, AI governance

1 Introduction

In September 2025, the DeepSeek team published a paper [1] in *Nature* introducing the DeepSeek-R1 inference model, which quickly garnered widespread attention from both academic and industrial communities. As the first research on a large language model (LLM) from China to be featured on the cover of *Nature*, it was described as the world's first peer-reviewed mainstream large language model. *Nature* noted in its coverage that “almost all mainstream large models have not yet undergone independent peer review, and this gap has finally been filled by DeepSeek.” The work presents notable achievements, including a technically ambitious “pure RL” training approach that forgoes supervised fine-tuning, a clear multistage pipeline for capability building and alignment, and



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rigorous evaluation across a comprehensive suite of reasoning and safety benchmarks. The public release of models and resources further enhances its practical utility for the research community.

While recognizing these promising contributions, the present analysis aims to constructively examine several critical limitations in the DeepSeek-R1 paper that merit further discussion. Our methodological approach is grounded in a structured analysis of the primary research article, its supplementary information, and publicly available peer review correspondence. We evaluate this work through the lens of established principles and frameworks for responsible and rigorous AI research—including “Datasheets for Datasets” [2], “Model Cards” [3], the NIST AI Risk Management Framework [4] and insights from foundational discussions on the societal impact of foundation models [5] and international AI principles [6]. These frameworks provide normative criteria for assessing transparency, reproducibility [7], safety, and ethical consideration, allowing us to systematically identify gaps between current reporting practices and emerging community standards. Our evaluation suggests that significant opportunities for improvement remain in meeting the high standards expected by premier scientific journals, particularly concerning data transparency, methodological completeness, and ethical oversight. These considerations carry significance beyond the scope of a single study. If not adequately addressed, such issues may over time erode the reliability and slow the cumulative progress of artificial intelligence research—a concern that becomes especially pressing as AI systems are increasingly applied in socially sensitive domains such as human health.

2 Methodological deficiencies

2.1 Data transparency and insufficient disclosure of data source

A primary concern regarding the methodological rigor of the DeepSeek-R1 research is the limited transparency surrounding its training data. While the paper states that the data “comes entirely from the internet,” this description lacks the specificity required for robust scientific evaluation. The internet encompasses information of widely varying quality, including biased, inaccurate, or copyrighted material. Without a clearer breakdown of data sources, composition, and collection methodologies—what constitutes true data transparency—it is challenging for the research community to assess the representativeness and potential biases inherent in the training corpus. This opacity, therefore, raises significant risks for the reproducibility of results and the critical evaluation of the model’s outputs [8]. Peer review documents indicate that reviewers sought more detailed information on data composition, yet the final publication remains notably vague on these points. This level of opacity stands in tension with widely recognized principles of scientific research, including the FAIR principles—Findable, Accessible, Interoperable, and Reusable—which serve as central guidelines for scientific data management [9]. To move beyond critique toward constructive remedy, Table 1 (Training Data Information Disclosure) presents a standardized template that operationalizes these transparency principles, specifying what constitutes meaningful data disclosure for rigorous scientific evaluation.

Table 1 Training data information disclosure

Information category	Specific content to be disclosed	Purpose and significance
Data sources and composition	<i>Specific dataset list:</i> Specify the public datasets used (such as C4, The Pile, corpora in specific languages or domains) and their versions <i>Description of self-collected data:</i> Describe the categories of seed websites crawled from the internet (such as news, academic, forums, code repositories), the time range, and the geographic/language distribution <i>Data proportion:</i> Provide the approximate mixing ratio of data from different sources or domains in the total training data	This allows researchers to assess data diversity, potential biases, domain coverage, and to understand the sources of the model's knowledge capabilities
Data preprocessing and cleaning	<i>Filtering rules:</i> Provide detailed explanations of filtering strategies and thresholds based on language, quality, toxicity, duplication, or privacy (such as the heuristic rules, classifiers used, and their precision/recall). <i>Deduplication methods:</i> Describe the deduplication methods and parameters at the document, paragraph, or substring level. Data cleaning has a significant impact on model safety and performance	Disclosing this information is crucial for reproducibility of results and also helps in assessing data quality
Data scale and statistics	<i>Number of tokens:</i> Total number of tokens after preprocessing, broken down by main language or domain. <i>Number of Documents/samples</i> <i>Key statistics:</i> such as word frequency distribution and sequence length distribution	Provides a basis for training efficiency, scaling law analysis, and fair comparison with other studies
Legal, ethical, and safety considerations	<i>Copyright and licensing status:</i> Describe the overall licensing framework of the dataset (e.g., public domain, open-source license), as well as statements regarding the handling of copy-righted materials. <i>Privacy handling:</i> Whether and how personally identifiable information (PII) is removed. <i>Harmful content:</i> Strategies for identifying and mitigating harmful content such as violence and hate speech.	Fulfill research ethics responsibilities, inform the community of potential legal and ethical risks, and promote responsible use
Data documentation and access	<i>Datasheets:</i> Provide standardized documentation following the "data sheet" framework proposed by Gebru et al. [2] <i>Access methods:</i> Even if the full dataset cannot be published due to size or copyright constraints, representative samples, data creation code, or detailed construction procedures should be provided	Datasheets are standardized tools for recording the motivation, composition, collection process, and maintenance of data, and are a central requirement in contemporary open science. Representative samples constitute the minimal requirement for verifying research claims

2.2 The challenge of training data contamination

The DeepSeek team acknowledges that its training corpus “may contain content generated by GPT-4,” characterizing this as unintentional contamination. However, the broader implications of this issue warrant further consideration. The potential presence of significant volumes of AI-generated content raises questions about the model's learned reasoning patterns and its claimed independence. Although the team reports implementing decontamination measures—such as the removal of approximately 6 million potentially contaminated mathematical samples—the specifics of the detection methodology and the efficacy of this process are not fully detailed [10]. Current techniques for reliably identifying and filtering AI-generated content remain an open research challenge, suggesting that complete eradication of contaminated samples is likely unattainable. Consequently, the possibility persists that the model's training and outputs are influenced by unidentified synthetic data, which could affect assessments of

its originality and self-contained reasoning capabilities [11]. This contamination challenge further underscores the need for systematic data provenance documentation—precisely the kind of transparency that the disclosure framework in Table 1 is designed to operationalize.

2.3 Model details and reproducibility challenges

Further limitations in methodological transparency relate to the disclosure of key model details, which presents obstacles for independent reproduction and verification. Although the paper provides an overview of the training process, critical specifics regarding architectural parameters, precise training protocols, and hyperparameter selection rationale are not comprehensively documented. As noted in the peer review feedback, decisions made during development often lack empirical justification, and the opaque nature of the training data limits reproducibility. This scarcity of detail also complicates a full assessment of the model's reported innovations. For example, the paper highlights a training cost of \$294,000 [12] but explicitly excludes the substantial pre-training costs of the base model (DeepSeek-V3 Base, estimated at ~\$6 million). Reporting costs in this selective manner may inadvertently give an incomplete picture of the total resources required for end-to-end model development, potentially misleading interpretations of the economic barriers to such research. These reproducibility challenges reinforce the importance of structured documentation practices—precisely the kind of systematic disclosure that the data transparency framework in Table 1 is designed to enable.

3 Ethics and safety risks

3.1 Challenges to academic integrity and intellectual property

The DeepSeek-R1 paper touches upon but does not sufficiently grapple with the profound challenges that large language models pose to academic integrity and intellectual property norms. As noted by scholars such as Liu and Jia, AI systems capable of synthesizing and generating academic-style text in real-time could lower barriers to potential misconduct, enabling the production of seemingly credible papers with minimal original human intellectual effort [13]. This capability not only risks facilitating plagiarism and fraud but also subtly challenges the human-centric model of knowledge production that underpins scientific progress. A further concern is the potential for models to generate inaccurate or misleading scholarly citations. While the DeepSeek-R1 paper references the use of external risk control systems, it offers limited detail on how such systems are designed to specifically prevent the generation of false or low-integrity academic content, effectively shifting the complex burden of ensuring factual and ethical output to downstream developers and users.

3.2 Assessments methodological limitations in safety evaluation

The research claims superior safety performance compared to other leading models based on standard benchmark evaluations. However, this safety assessment is subject to important methodological constraints. Firstly, static safety benchmarks may not encompass the full spectrum of novel, adaptive malicious uses a deployed model might encounter. Secondly, evaluations focused on known attack patterns may not reliably predict resilience against unforeseen manipulation techniques. A critical practical gap lies

in the significant difference between the safety of the openly released model and that of a version integrated with a proprietary, external risk control system. This discrepancy suggests that most real-world implementations, which rely on the open-source version, could operate with higher latent risk. The paper’s recommendation for developers to implement their own safeguards, while pragmatic, is not accompanied by concrete implementation guidelines or validated frameworks, leaving a gap between principle and practice in responsible deployment. To address this gap, Table 2 (Dimensions of AI Governance Responsibility) provides a multi-dimensional framework that operationalizes the core responsibilities—transparency, hazard prevention, stakeholder communication, accountability, and ethical compliance—offering a systematic lens for translating abstract principles into actionable evaluation criteria.

3.3 Attribution of responsibility and accountability mechanism

The study does not adequately address the unresolved question of liability and accountability for content generated by models like DeepSeek-R1. In high-stakes domains such as healthcare, reliance on a model whose training data biases are not fully transparent, or whose safety assurances may be overstated, could lead to harmful outcomes—from diagnostic errors to biased recommendations. Current legal and professional frameworks struggle to attribute responsibility across the chain of developers, deployers, and end-users when harm occurs, a challenge amplified by the “black box” nature of complex AI systems. This accountability vacuum is particularly acute in academia regarding authorship. As Liu and Jia highlight, copyright law typically requires a legal person or entity that can claim ownership and bear liability [13]. AI systems themselves cannot hold copyright or be held legally responsible for infringement or plagiarism, creating a

Table 2 Dimensions of AI governance responsibility

Responsibility dimension	Specific connotation (manifestation in AI safety research) potential	Requirements for the Deepseek-R1 case
Transparency and reproducibility	Provide detailed disclosure of research methods, data, and limitations, enabling other researchers to assess and replicate the results. (See FAIR Principles)	Not only should safety benchmark scores be published, but the composition of the test set, evaluation methods, known blind spots, and the specific architecture or proxy indicators of external risk control systems should also be explained in detail
Hazard prevention and risk warning	Proactively identify and warn about the risks of research results being maliciously misused, and propose mitigation measures. (See EU AI Act)	Clearly warn about the security gap between the open-source version and the commercial version with a full risk control system, and quantitatively explain the level of risk that this gap may cause
Stakeholder communication	Clearly communicate the capability boundaries and risks of the model to users, developers, and the general public. (See the ACM Code of Ethics and Professional Conduct)	In prominent places such as papers and model release pages, explain safety assumptions and scopes of applicability in non-technical language, avoiding statements like ‘safety leading’ that could mislead non-professional users
Accountability	Take certain responsibility for the consequences of research, providing channels for follow-up improvements and issue feedback. (See the OECD AI Principles and NIST AI RMF)	Provide channels for disclosing security vulnerabilities, and for the suggestion of ‘advising developers to deploy similar mechanisms,’ offer a basic framework, evaluation tools, or best practice guidelines, not just a statement
Comply with ethical standards	Adhere to general research ethics and AI ethical principles, such as fairness, non-harm, and privacy protection. (See UNESCO’s Recommendation on the Ethics of AI)	Ensure the diversity and impartiality of safety evaluation datasets, and assess whether safety interventions may introduce new forms of unfairness or discrimination

Table 3 Standards for the social responsibility of academic research

Recommendation	Measures	Primary responsible party	Action
Enhance research transparency	Promote the use of structured documentation (data sheets, model cards) and reproducibility checklists	Editorial boards and program committees of top academic conferences and journals (such as NeurIPS, ICML, and the Nature series)	Revise the submission guidelines to make the submission of structured documents and the completion of reproducibility checklists mandatory requirements for paper acceptance
Strengthening ethics and impact assessment	Introduce interdisciplinary preliminary review or dedicated review procedures for AI research with significant social impact	Academic publishers leading the formation of reviewer pools, funding agencies such as the National Science Foundation, and large corporate research institutes with internal ethics review committees should participate	Establish a permanent pool of reviewers encompassing experts in law, ethics, and sociology; require project completion reports to include a section on social impact assessment
Establish independent verification and auditing	Develop an ecosystem for third-party technical verification and security auditing	National standards and technical organizations (such as NIST), professional associations (such as ACM, IEEE), and accredited non-profit research laboratories	Lead the development of benchmarking and audit standard frameworks; organize and implement independent performance and security audits for major model releases
Clarify the framework of legal rights and responsibilities	Define the copyright ownership and infringement liability chain for AI-generated content	National legislative bodies and judicial systems leading legal amendments and judicial interpretations	Amend the Copyright Law and related regulations to specify the status of generated content as a work when it meets the ‘originality’ criterion, define the boundaries of ‘fair use’ for training data, establish developers as the primary party responsible for data compliance, and uphold that users bear ultimate responsibility for the output content

fundamental mismatch between the technology’s capabilities and existing systems for assigning rights and responsibilities for generated text. The DeepSeek-R1 paper leaves this critical governance challenge largely unexamined. This accountability vacuum directly points to the need for clearer legal frameworks—precisely the kind of governance clarity that the “legal rights and responsibilities” recommendation in Table 3 is designed to advance.

4 Reassessing scientific contributions

4.1 Scrutinizing claims of innovation

The central innovation claim of the DeepSeek-R1 paper is its demonstration that “reinforcement learning alone can teach large language models to reason,” purportedly without extensive human data or prompt engineering. This proposition merits careful contextualization. The application of reinforcement learning (RL) to align language models with human preferences is an established research direction, as evidenced by prior work from other leading labs. While DeepSeek-R1’s specific implementation—particularly its “pure RL” approach and techniques like group relative policy optimization—represents a noteworthy and ambitious methodological choice, its characterization as a foundational breakthrough may be overly broad. A more precise assessment would require a detailed comparative analysis against existing RL frameworks to delineate its unique contributions and incremental advances. Furthermore, the theoretical

interpretation of how RL incentivizes “reasoning” could be refined; the process is perhaps more accurately described as optimizing model outputs to satisfy externally defined reward functions that correlate with successful outcomes, rather than directly instilling an internal reasoning faculty. The peer review process itself noted that several design decisions lacked accompanying empirical justification, suggesting room for more thorough methodological validation.

4.2 About low-cost training

The paper highlights a training cost of approximately \$294,000, a figure that presents a notably efficient economic profile. However, a full understanding of this cost requires important contextualization. This amount specifically pertains to the reinforcement learning fine-tuning phase and explicitly excludes the substantial pre-training investment required for the base model, DeepSeek-V3 Base, whose development cost is estimated in the millions of dollars. Presenting the fine-tuning cost in isolation, while technically accurate, provides an incomplete picture of the total resource expenditure necessary for end-to-end model development. A comprehensive cost accounting would also consider related but often uncatalyzed expenditures, such as data acquisition, curation, human expertise, and sustained infrastructure. Providing this broader context is crucial to inform accurate perceptions among researchers and policymakers regarding the true economic thresholds and resource requirements of state-of-the-art AI research.

4.3 Limitations of the peer review process

Nature’s presentation of DeepSeek-R1 as the “world’s first peer-reviewed mainstream large language model” underscores a significant step toward formal academic validation in the field. Nevertheless, this process invites reflection on its scope and efficacy. The dedicated review period, while resulting in a detailed 64-page review report, spanned a relatively condensed timeframe for a work of such technical and societal complexity. More notably, the reviewer composition, as disclosed, appears to have been predominantly from core machine learning and technical domains. The inclusion of perspectives from the humanities and social sciences—such as ethics, law, and science and technology studies—could have enriched the evaluation, particularly for assessing the paper’s discussion of societal implications and ethical considerations. A more diverse review panel might have more rigorously probed the translation of technical claims into real-world impact, aligning with broader principles for responsible and human-centric AI development that extend beyond pure technical performance. This observation highlights a potential area for evolution in peer-review practices for high-impact, transformative AI research.

5 Discussion and conclusion

This critical analysis has identified several interconnected limitations in the DeepSeek-R1 paper pertaining to data transparency, methodological reproducibility, safety evaluation, and the assessment of broader societal implications. While the study represents a significant technical achievement, these shortcomings highlight areas where the norms of high-impact scientific publishing can be strengthened to better meet the unique challenges posed by frontier AI research. The issues examined—such as opaque data provenance, undefined accountability for generated content, and peer review focused

primarily on technical merit—are not unique to this single paper but reflect systemic tensions in a rapidly evolving field. If left unaddressed, they risk eroding the credibility of AI research and hindering its responsible integration into society.

To translate this critique into constructive pathways for improvement, we direct actionable recommendations to the specific stakeholders best positioned to enact change. These proposals are designed to form a coherent, multi-tiered framework for enhancing rigor and responsibility. To visualize how these proposals cohere into an integrated action agenda, Table 3 presents a structured framework for enhancing the social responsibility of academic AI research.

The framework in Table 3 underscores a central insight of our analysis: responsible AI research is not the responsibility of any single actor but requires coordinated action across the entire ecosystem. The four recommendations outlined in the table—Transparency, Ethics Assessment, Independent Auditing, and Legal Clarity—are interdependent. For instance, mandatory disclosure of data (Recommendation 1) provides the necessary foundation for independent auditors (Recommendation 3) to verify safety claims, while legal frameworks (Recommendation 4) define the accountability that incentivizes both transparency and rigorous ethics assessment (Recommendation 2). The DeepSeek-R1 case illuminates both the necessity and the current absence of such coordinated action. In the following sections, we elaborate on how each of these four pillars can be operationalized by their respective stakeholders, beginning with the role of the academic publishing community.

First, the academic publishing community must take the lead in institutionalizing transparency and broadening evaluative criteria. This directly operationalizes the first recommendation in Table 3. Building on existing initiatives like NeurIPS, leading conferences and journals should mandate structured documentation—such as datasheets for datasets and model cards—as a non-negotiable component of the submission and publication process [14]. This would formalize the disclosure of data composition, collection methods, and known biases, directly addressing the reproducibility challenges identified in our analysis. Furthermore, for research presenting models with clear societal impact, these publishing bodies should pilot interdisciplinary review panels that include scholars from the humanities and social sciences. This would ensure that evaluations of technological capability are systematically integrated with considerations of ethical alignment and potential societal consequences, thereby raising the standard for what constitutes “high-quality” AI research.

Secondly, the establishment of an independent verification ecosystem is crucial to validate claims and ensure community trust, corresponding to the third recommendation in Table 3. The concerns regarding benchmarking limitations and potential overstatement of capabilities underscore the need for objective, third-party assessment. This requires collaboration across different actors: national standards bodies (e.g., NIST) can develop robust evaluation benchmarks and audit frameworks; professional societies (e.g., ACM, IEEE) and accredited research institutes can perform the actual audits of major models. The creation of this “quality assurance” layer would provide a trusted mechanism to verify performance and safety claims, benefiting both the research community and downstream adopters by reducing information asymmetry and hype.

Thirdly, policymakers and regulators play an indispensable role in creating the stable governance frameworks necessary for long-term, responsible innovation, directly

addressing the fourth recommendation in Table 3. The legal and accountability vacuums surrounding AI-generated content, as discussed, cannot be solved by technical or procedural measures alone. Legislators and regulatory agencies must therefore advance efforts—informed by initiatives like the EU AI Act—to define clear standards for accountability, provenance, and safety in high-risk applications [15]. Providing this legal clarity is not a constraint but a necessary foundation that enables researchers and developers to innovate with a clearer understanding of their responsibilities and the guardrails that protect public interest. Building upon the EU's approach as an important reference point for comprehensive AI governance, it is instructive to examine how major AI-developing nations are constructing their own regulatory ecosystems—with China offering a particularly illustrative case of rapid framework development that combines cybersecurity fundamentals with AI-specific requirements.

To provide concrete regulatory context, China has established a multi-layered framework governing AI system. The Cybersecurity Law (2017) and Data Security Law (2021) establish foundational obligations for data handling and national security protection. Building on these, the Interim Measures for the Management of Generative Artificial Intelligence Services (2023) specifically target generative AI, requiring service providers to: (a) ensure training data legality and compliance with intellectual property rights; (b) implement measures to prevent algorithmic discrimination; (c) fulfill personal information protection duties; and (d) conduct security assessments and register algorithms with the Cyberspace Administration for services with “public opinion attributes.” Technical standards such as TC260-003 (Security Basic Requirements for Generative AI Services) further operationalize these principles through quantitative metrics, including $\geq 95\%$ rejection rates for harmful content and specific content filtering requirements. Yet even with this robust domestic framework in place, the DeepSeek-R1 case reveals that the governance challenge extends far beyond any single jurisdiction's regulatory capacity.

The challenges illuminated by the DeepSeek-R1 case are not merely a reflection of a domestic regulatory gap, but rather a broader global governance dilemma: how to ensure safety without stifling innovation. The EU AI Act's recent pivot—withdrawing the AI Liability Directive amid concerns about over-regulation—adds an economic dimension to this common dilemma [16].

In summary, the journey toward truly rigorous and socially responsible AI research requires deliberate, collaborative action across academia, industry, and policy. The DeepSeek-R1 publication serves as a catalyst for this necessary discussion, reminding us that the scientific and ethical maturity of the field must advance in lockstep with its technical ambitions. By adopting these stakeholder-specific recommendations, the community can build a more robust, transparent, and trustworthy ecosystem for the development and evaluation of transformative AI technologies.

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Huimin Peng conceived the topic and drafted the text.

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Data availability

No datasets were generated or analysed during the current study.

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